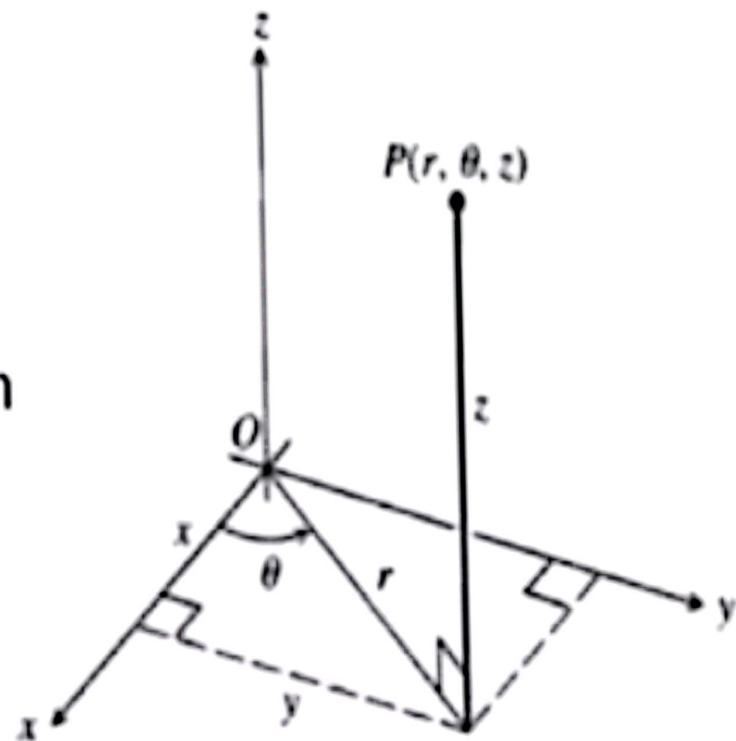


Cylindrical Coordinates

Cylindrical coordinates for space can be obtained by combining polar coordinates in the xy -plane with the usual z -axis. This assigns to every point in space one or more coordinate triples of the form

(r, θ, z) . Cylindrical coordinate represent a point P in space by ordered triples (r, θ, z) in which

- 1) r and θ are polar coordinates for the vertical projection of P on the xy -plane,
- 2) z is the rectangular vertical coordinate. -



Equation Relating Rectangular (x, y, z) and Cylindrical (r, θ, z) Coordinates

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

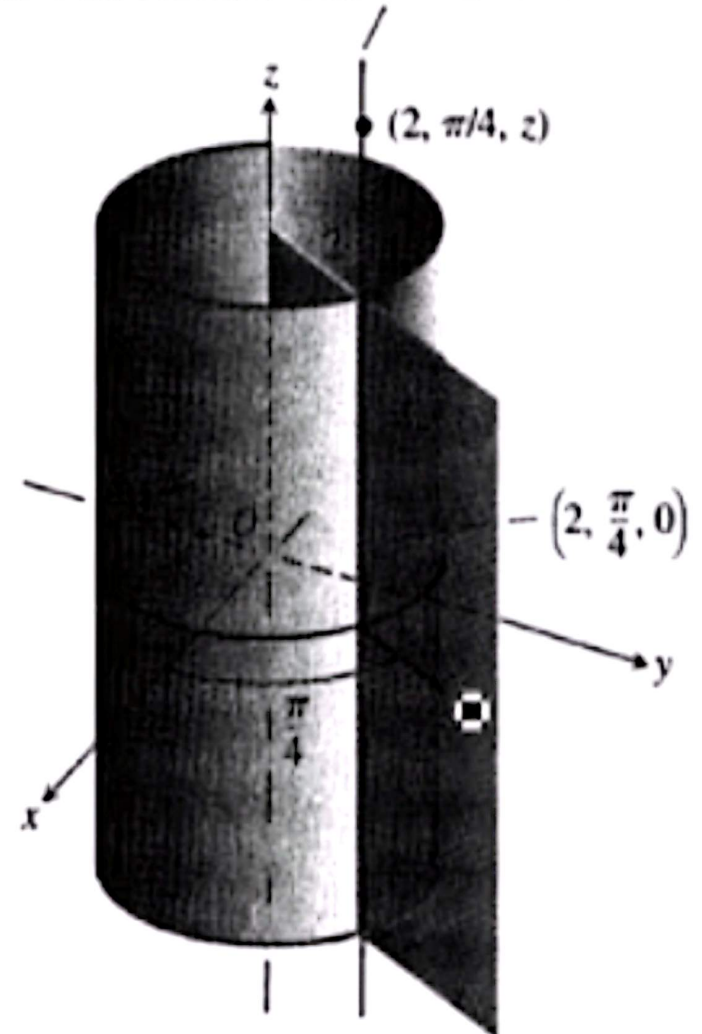
$$r^2 = x^2 + y^2, \quad \tan \theta = \frac{y}{x}$$

In cylindrical coordinates, the equation $r = a$ describes not just a circle in the xy -plane but an entire cylinder about the z -axis.

Example # 1 What points satisfy the equation

$$r = 2, \quad \theta = \frac{\pi}{4}?$$

Along this line, z varies while r and θ have the constant values $r = 2$ and $\theta = \pi/4$.



Example # 2 Sketch the surface $r = 1 + \cos \theta$.

$$r = 1 + \cos \theta \quad z = z$$

$$\theta = 0 \Rightarrow r = 2$$

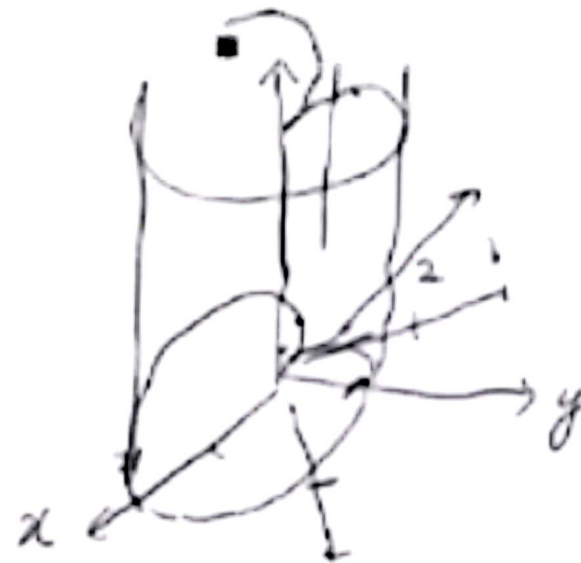
$$\theta = \frac{\pi}{6} \Rightarrow r = 1 + \frac{\sqrt{3}}{2} = 1.9$$

$$\theta = \frac{\pi}{4} \Rightarrow r = 1 + \frac{1}{\sqrt{2}} = 1.7$$

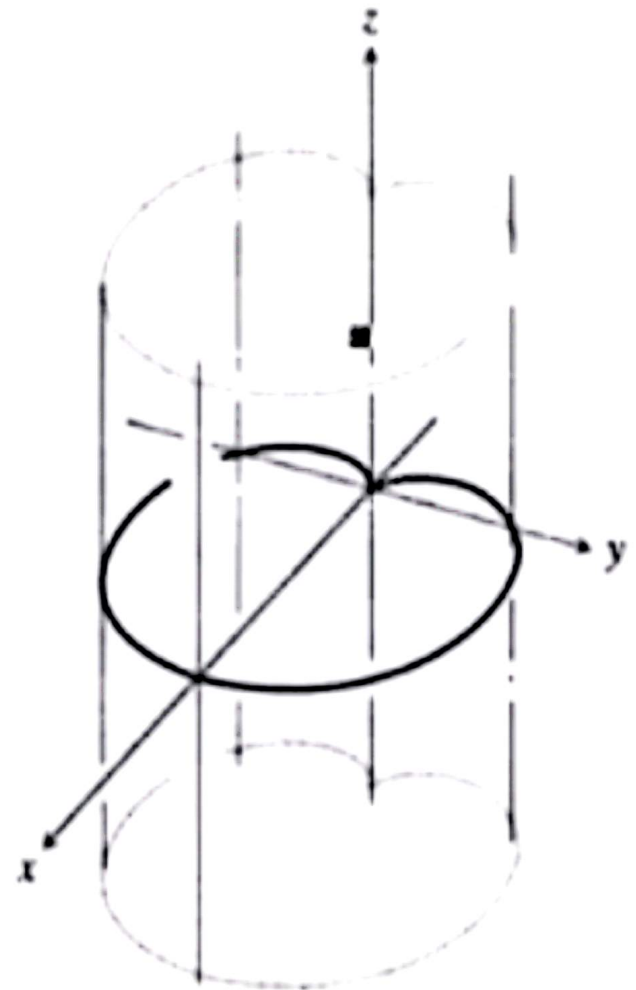
$$\theta = \frac{\pi}{3} \Rightarrow r = 1 + \frac{1}{2} = 1.5$$

$$\theta = \frac{\pi}{2} \Rightarrow r = 1$$

$$\theta = \pi \Rightarrow r = 0$$



Example # 2 Sketch the surface $r = 1 + \cos \theta$.



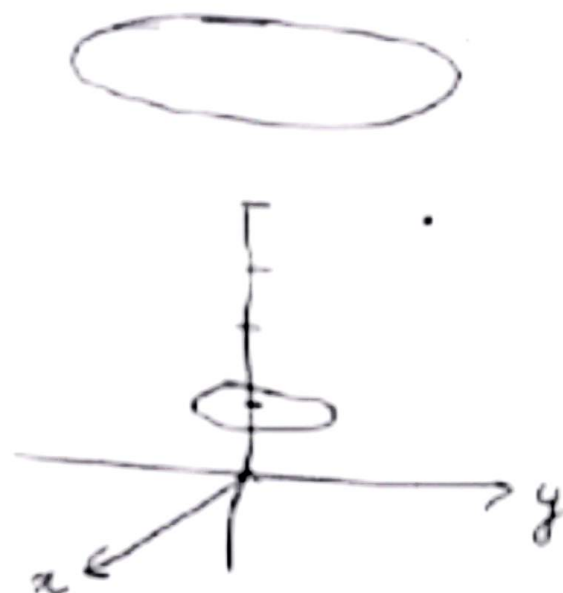
Example # 3 Find a Cartesian equation for the surface $z = r^2$ and identify the surface.

$$z = r^2$$

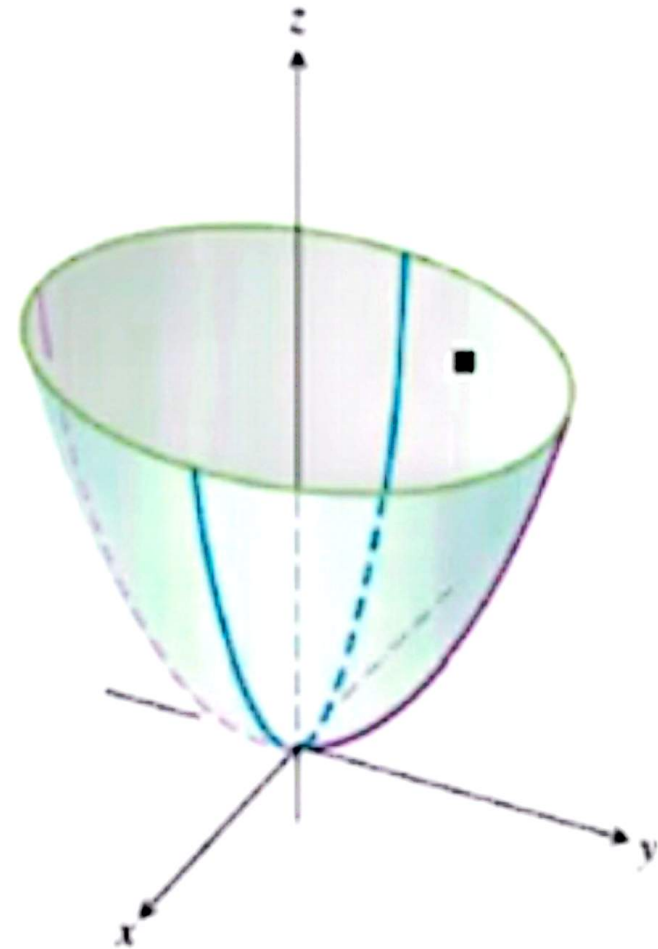
$$r^2 = x^2 + y^2$$

$$z = x^2 + y^2$$

r	0	± 1	± 2	± 3
z	0	1	4	9



Example # 3 Find a Cartesian equation for the surface $z = r^2$ and identify the surface.



Example # 4 Find an equation for the circular cylinder $4x^2 + 4y^2 = 9$ in the cylindrical coordinates.

$$4x^2 + 4y^2 = 9$$

$$4(x^2 + y^2) = 9$$

$$4r^2 = 9$$

$$r^2 = \frac{9}{4}$$

$$r = \pm \frac{3}{2}$$

■

Example # 5 Find an equation for the cylinder $x^2 + (y - 3)^2 = 9$ in the cylindrical coordinates.

$$x^2 + (y - 3)^2 = 9$$

$$x^2 + y^2 - 6y + 9 = 9$$

$$x^2 + y^2 - 6y = 0$$

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r^2 = x^2 + y^2$$

$$x^2 + y^2 - 6y = 0$$

$$r^2 - 6r \sin \theta = 0$$

$$r(r - 6 \sin \theta) = 0$$

$$r = 6 \sin \theta$$

Cylinder with center $(0, 3, z)$,
radius of the cylinder = 3.

Example # 5 Find an equation for the cylinder $x^2 + (y - 3)^2 = 9$ in the cylindrical coordinates.

